

Duncan Austin

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OVERVIEW

My current research is as a member of the PEARLS and VENUS collaborations, searching and classifying high-redshift galaxies in deep optical and near-infrared data from both the Hubble Space Telescope (HST) and James Webb Space Telescope (JWST). In particular I am interested in the physics governing ionizing photon production and escape from the ISM, which has profound implications for the reionization history and topology. During my academic studies as a PhD student and post-doctoral researcher at the University of Manchester I have developed a highly modular and public cataloguing software, *galfind*, for galaxy identification, classification, and sample selection. Using this photometric toolbox, I have identified several new high-redshift galaxy candidates from the first billion years of the Universe's history using state-of-the art JWST/NIRCam photometric imaging, leading to the development of two high-redshift paper series'.

EDUCATION

- **University of Manchester** 2025 - Onwards
PDRA - ERC funding provided by Chris Conselice Manchester, United Kingdom
 - Research interests: Reionization, UV luminosity functions, SED fitting, cataloguing and data reduction techniques for JWST/NIRCam, galaxy clustering and cosmic variance modelling
- **University of Manchester** 2021 - 2025
PhD Astrophysics - STFC funded Manchester, United Kingdom
 - Thesis title: [Inferring the properties of star forming galaxies in the Epoch of Reionization with JWST](#)
- **University of Manchester** 2017 - 2021
MPhys (Hons) - 1st class Manchester, United Kingdom
 - Received outstanding achievement letter for obtaining an average grade > 80%, signed by Chief Examiner and Assessment Lead, Judith McGovern

RESEARCH SUMMARY

- **EPOCHS-DR2 II: Blank field UV luminosity functions highlight an elevated and evolving star formation efficiency** December 2025 - Ongoing
PDRA, University of Manchester (paper to be submitted within ~ 1 – 2 months)
 - UVLFs calculated using unique $1/V_{\text{max}}$ technique that incorporates non-uniform field depths to remove bias towards flatter faint end slopes found in regular studies
 - HOD framework used to estimate halo masses and provide contours in the UV luminosity–halo mass scatter vs star formation efficiency parameter space from joint UVLF + galaxy bias constraints
- **EPOCHS-DR2 I: Expanded data release and properties of $6.5 < z < 16.5$ galaxies across NIRCam deep fields covering $\sim 700 \text{ arcmin}^2$** December 2025 - Ongoing
PDRA, University of Manchester (paper in final internal review stage)
 - Presents fiducial sample of $\sim 2,500$ galaxy candidates at $6.5 < z < 16.5$ from HST/ACS_WFC and JWST/NIRCam data. Cross matched to 292 NIRSpec/PRISM spectra from the DJA v4.4
 - Most comprehensive completeness/contamination estimates in the EoR to date using real spectra

- KL-divergence analysis performed on the output Bagpipes SED fitting results, highlighting which parameters/hyperparameters are not adequately constrained as a function of redshift, including $f_{\text{esc}}^{\text{LyC}}$
 - Includes “gold” sample of $z > 10.5$ candidates with an increased SNR detection threshold. The estimated 60 – 75% contamination rate suggests that 5 – 7 should be real high-redshift objects, in minor tension with McLeod+2026 and Weibel+2026. Two blue $z \simeq 14$ candidates have robust $> 3\sigma$ F335M and F410M detections which merit single-slit NIRSpec follow-up
- **The galfind optical/NIR photometric toolbox** September 2022 - Ongoing
PhD, University of Manchester [ galfind]
- Includes cataloguing tool to perform source extraction, depth calculation, automasking, sample selection, SED fitting, photometric property calculation, luminosity / mass functions, and morphological fitting
 - Written in a flexible object oriented framework, including ReadTheDocs documentation
- **Resolving the ionizing photon budget crisis with HII clumping constraints at $z \simeq 6$** July 2024 - July 2026
PhD, University of Manchester
- Calculates ionizing photon production rates and efficiencies as well as Lyman continuum escape fractions for a large sample of star forming and smouldering galaxies across a range of deep, blank field, photometric surveys
 - Clumping factor constraints required to match direct x_{HI} measurements, solving the ionizing photon budget crisis
- **UV continuum slopes from wideband JWST/NIRCam photometry** July 2023 - October 2025
PhD, University of Manchester
- Used the galfind software to produce a sample of > 1000 star forming galaxies at $6.5 < z < 13$ (the EPOCHS series)
 - Thorough analysis to outline (and partially correct) for UV continuum slope, β , biases associated with underestimated $\text{Ly}\alpha$ emission and damped $\text{Ly}\alpha$ absorption, as well as strong rest-frame UV emission lines
 - Reveals lack of dust in star-forming galaxies at high redshift, in tension with simulations
- **High-redshift SFGs in the NGDEEP survey** February 2023 - July 2023
PhD, University of Manchester
- Performed source extraction, deblending, and forced photometry with SExtractor using a variety of photometric filters for selection
 - Calculated photometric redshifts using appropriate SFG templates within EAZY-py, and constructed a robust sample including various redshift PDF and SNR criteria
- **Galaxy clustering analysis in the UDS field** September 2021 - September 2022
PhD, University of Manchester
- Calculated galaxy biases and halo masses at $z < 4$ for a large sample of $\sim 10,000$ Lyman break galaxies over 0.8 deg^2 to investigate the redshift dependence of the angular galaxy clustering signal
 - Observed halo downsizing by performing analysis in a variety of stellar mass bins
- **Scale dependence of the linear HI bias approximation in semi-analytical models** September 2020 - July 2021
MPhys, University of Manchester
- Used the L-Galaxies, GAEA, and DARK SAGE semi-analytical models (SAMs) to explore baryonic feedback and cold gas partitioning via the HI mass function, HI-stellar, and HI-halo mass relations
 - Computed a basic HI power spectrum and investigated the scale dependence of the HI bias to determine the appropriateness of the linear bias approximation in the case of each SAM

RESEARCH ACTIVITY

Conferences

- **The many scales of galaxy environments** July 2026
University of Durham, Ascona Switzerland 
 - Presentation: “Implications for IGM clumping and reionization from H α emitters at $z \simeq 6$ ”
 - This included highlights from [Li+2026](#): “Reionization in protocluster environments at $z > 7$ with JWST/NIRSpec”
- **Manchester–Liverpool–Lancaster Extragalactic Meeting** May 2026
University of Manchester
 - Presentation: “EPOCHS-DR2: UVLFs, SFR density and cosmic variance estimates at $6.5 < z < 13.5$ with JWST/NIRCam”
- **Miracles of the Early Universe** June 2025
University of Geneva
 - Presentation: “Resolving the ionizing photon budget crisis with HII clumping constraints at $z \simeq 6$ ”
- **Cosmic Dawn at High Latitudes** June 2024
University of Stockholm 
 - Presentation: “Unbiased UV continuum slopes at $z = 6.5 - 13$ from combined PEARLS GTO and public JWST NIRCam imaging”
- **Semaine de L’astrophysique Francaise (SF2A)** June 2024
L’aboratoire d’Astrophysique de Marseille 
 - Presentation: “Unbiased UV continuum slopes at $6.5 < z < 13$ from the EPOCHS sample”
- **The First Year of JWST Science** September 2023
Space Telescope Science Institute (STScI), Baltimore 
 - Presentation: “UV continuum slopes and their redshift evolution with PEARLS GTO and public JWST NIRCam imaging”
- **National Astronomy Meeting** July 2023
University of Cardiff 
 - Presentation: “UV continuum slopes and SFRs from combined PEARLS GTO and public JWST NIRCam imaging”

TEACHING AND LEADERSHIP EXPERIENCE

- **Masters student supervision** September 2023 - Present
University of Manchester
 - Supervisory aid to several MScR and MPhys students, including Honor Harris, Caio Goolsby, James Arcidiacono, Aimee Parker, Giorgia Riley, Harry Thomas, Joe Hopkins, Harry Whittingham, Oliver Rogers, Julien Marabotto, and Albert Turner-Gale
 - Contribution to project planning via Slack and regular in-person/online meetings
- **1st year undergraduate laboratory teaching** September 2021 - April 2022
University of Manchester
 - Teaching of measurement and data analysis techniques, as well as the propagation of uncorrelated uncertainties
 - Marking of lab reports and discussion based interviews for the stellar astrophysics and e/m ratio experiments

SCIENTIFIC OUTREACH AND PUBLIC ENGAGEMENT

- **Science demonstrator**

July 2022, July 2023

Bluedot festival



- Voluntary work for both the JBCA and JWST extragalactic astrophysics stands
- Discussion of astrophysical principles to inspire the general public, including children. Hands-on demonstrations include VR headsets displaying the multi-wavelength night sky, an IR camera to showcase JWST/NIRCam photometry, and spectroscopy with fluid tubes using a diffraction grating

ADDITIONAL SKILLS

- **Programming Languages:** Python, C++ (both object oriented and functional)
- **Software:** SAOImage DS9, TOPCAT, SExtractor, Photutils, EAZY-py, Bagpipes, GALFIT, pysersic, msaexp
- **Mathematical & Statistical Techniques:** Bootstrapping, 2-pt clustering statistics, Bayesian inference and Monte-Carlo Markov Chain sampling
- **DevOps & Version Control:** Continuous Integration (CI) with GitHub Actions, Sphinx + ReadTheDocs, pytest, ruff, pre-commit

REFERENCES

1. **Christopher, J. Conselice**

Professor of Extragalactic Astronomy, Jodrell Bank Centre for Astrophysics
University of Manchester
Email: conselice@manchester.ac.uk
Relationship: PhD Supervisor

2. **Laura Wolz**

Presidential Fellow, Jodrell Bank Centre for Astrophysics
University of Manchester
Email: laura.wolz@manchester.ac.uk
Relationship: Masters project supervisor and internal PhD examiner